Machine Learning Mini Project
UE24CS352A

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TITLE :- DETECTING AND PREDICTING EARTHQUAKE GROUNDMOTION DIRECTIONALITY

ABSTRACT

Ground-motion directionality describes the horizontal orientation associated with the maximum response recorded during an earthquake. This project follows the research problem of predicting the absolute angular difference between the maximum-response orientations at two sites affected by the same earthquake. Using PEER NGA-West2 acceleration records and flatfile metadata, the workflow parses waveforms, estimates orientations, constructs same-event site pairs, and compares regression models. The available records support an exploratory analysis; the strict 5 km subset is too small for reliable model evaluation, so a broader 100 km experiment is reported separately.

1. Problem Statement & Objective

The objective is to investigate whether the angular difference between the maximum-response orientations of two recording sites can be predicted from their separation distance and available earthquake and site attributes. The research is motivated by evidence that nearby sites may exhibit related directionality patterns, rather than behaving as if their orientations were entirely independent.

2. Dataset and Preparation

The project uses a limited, non-random subset of PEER NGA-West2 waveform records and corresponding flatfile metadata. Of 100 downloaded record sequence numbers (RSNs), 93 records were usable and matched to metadata. The broader same-event pairing procedure produced 269 pairs at separations of up to 100 km, spanning nine earthquakes. Applying the paper's 5 km criterion left eight pairs across five earthquakes. The flatfile provides candidate event and site attributes, while the directionality target is derived from the acceleration waveforms.

3. Methodology

• Waveform processing: read acceleration time series and sampling interval from AT2 files and identify the two horizontal components.
• Orientation estimation: combine the horizontal components over candidate orientations and estimate the orientation associated with maximum response.
• Target and pair construction: pair records from the same earthquake and compute the absolute angular difference between the two site orientations, using the project's orientation convention.
• Feature preparation: join waveform-derived information with NGA-West2 metadata and construct pair-level predictors from the available event/site fields.
• Model comparison: compare a distance-only linear benchmark, Ridge regression and a multilayer perceptron (MLP), using R^2 and mean absolute error (MAE) to describe predictive performance.

4. Implementation Overview

The workflow is implemented in Python and organized into modules for record parsing, waveform processing, metadata matching, pair construction, model fitting and pipeline execution. Supporting Jupyter notebooks cover directionality fundamentals, rotation-mathematics validation and NGA-West2 flatfile inspection. The main scripts generate processed CSV/JSON outputs and plots, while the Google Colab notebook and setup instructions document execution. Raw waveform data and the NGA-West2 flatfile are not included in the public repository.

5. Results

The strict 5 km subset contains too few pairs to support a reliable held-out model comparison. The following table reports the separate exploratory experiment using pairs within 100 km and a random example-level 80/10/10 split:

Model	Test MAE (°)	R^2
Distance-only linear benchmark	22.70	–0.073
Ridge regression (α = 1.5)	22.91	–0.118
Validation-selected MLP	23.15	–0.138

Negative R^2 values indicate that the models did not outperform predicting the test-set mean under this split. The distance-only baseline slightly outperformed Ridge and MLP in this supplementary experiment. A separate event-held-out evaluation was also unstable, with negative R^2 values and a small event-level test set. These findings should be interpreted as exploratory and are not directly comparable with the paper's 5 km experiment.

6. Interpretation and Significance

The project demonstrates an end-to-end workflow from waveform parsing to orientation estimation, site-pair construction and regression comparison. However, the limited and non-random waveform subset restricts the strength of the conclusions. The broader 100 km pairing rule changes the spatial scope of the analysis, and the available predictor set has not been verified as identical to the paper's final selected features. Consequently, the experiments are useful for validating the workflow and identifying data requirements, but they do not establish reliable predictive performance under the paper's original setting.

7. Conclusion

This project implements a paper-informed machine-learning pipeline for investigating differences in earthquake ground-motion directionality between recording sites. It combines waveform-derived orientation estimates with earthquake and site metadata, constructs same-event pairs, and compares a distance-only baseline with Ridge regression and an MLP. On the available supplementary 100 km split, the machine-learning models did not demonstrate a clear improvement over the simple baseline. Future work should expand the usable waveform set, verify component azimuth conventions, align feature selection more closely with the proposed study, and evaluate models using event-aware splits once a sufficiently large 5 km dataset is available. The current analysis is exploratory and is not intended for real-time earthquake prediction or engineering decisions.

Reference

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4. Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

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